

Alexandria University
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HVDC Transmission System

Simulink VSC Assignment

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Submitted to:

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Simulink Model Link:
Click here to access the Simulink Model on GitHub

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Abstract

This report presents the mathematical modeling and simulation of a grid-connected Two-Level Voltage Source Converter (VSC). The system utilizes a Hysteresis Current Controller operating in the *abc* reference frame to achieve decoupled, independent regulation of active and reactive power. The dynamic performance and stability of the VSC are evaluated across five distinct operational scenarios, including pure active power injection, symmetrical power delivery, STATCOM operation, power reversal (rectification), and a severe transient step change. The results show reference tracking in all five cases, with the power reversal completed within a fraction of an AC cycle.

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1 System Structure and Overview

The implemented model consists of a grid-connected Two-Level Voltage Source Converter (VSC), which is a fundamental building block in modern High Voltage Direct Current (HVDC) transmission systems. Unlike traditional Line-Commutated Converters (LCC), VSCs offer the significant advantage of independent regulation of both active (P) and reactive (Q) power.

The primary objective of this simulation is to validate the capability of the VSC to inject into or absorb power from a stiff 90 kV (peak), 50 Hz AC grid. The converter is connected to the grid through a coupling inductor (represented by an $R - L$ branch). This inductive filter is crucial as it suppresses the high-frequency switching harmonics generated by the converter switching, which reduces the high-frequency content of the injected currents. The DC side is supplied by a split-capacitor DC link to establish a stable reference voltage for the two-level switching.

1.1 System Specifications

The key parameters utilized for the design and simulation of the VSC system are summarized in Table 1.

Table 1: VSC System Simulation Parameters

Parameter	**Value**
AC Grid Peak Phase Voltage (V_m)	90 kV
System Frequency (f)	50 Hz
Maximum Active Power (P)	$\pm$405 MW
Maximum Reactive Power (Q)	$\pm$405 MVAR
Control Strategy	Hysteresis Current Control
Reference Frame	*abc* frame

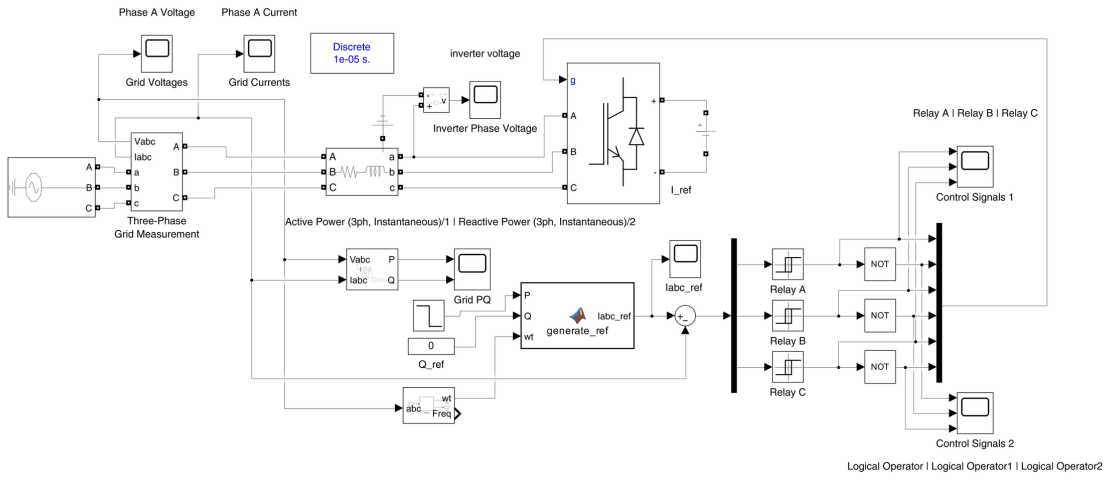


Figure 1: Complete Simulink Model of the Two-Level VSC System.

2 Control Strategy

The control strategy relies on a Hysteresis Current Controller operating directly in the *abc* reference frame. This non-linear feedback control method is renowned for its outstanding dynamic response, unconditional stability, and simplicity in implementation, as it avoids complex inner current loop PI tuning and coordinate transformations ($dq0$).

2.1 Reference Current Generation

The reference currents (I^*_{abc}) are generated analytically based on the instantaneous required active and reactive power values. The theoretical magnitude of the reference phase current is calculated based on the apparent power equation:

$$S = \sqrt{P^2 + Q^2}, \quad S = \frac{3}{2} V_m I_m \implies I_m = \frac{2}{3} \frac{S}{V_m} \tag{1}$$

Where V_m is the peak phase voltage (90 kV). The power factor angle, defining the phase shift between the grid voltage and the injected current, is determined by $\phi = \tan^{-1}(Q/P)$. The following MATLAB function was utilized in the simulation to continuously generate the instantaneous synchronized reference signals:

```
function Iabc_ref = generate_ref(P, Q, wt)
    Vm = 90000; % Peak phase voltage

    % Calculate peak current
    S = sqrt(P^2 + Q^2);
    Im = (2/3) * (S / Vm);

    % Calculate power factor angle
    phi = atan2(Q, P);

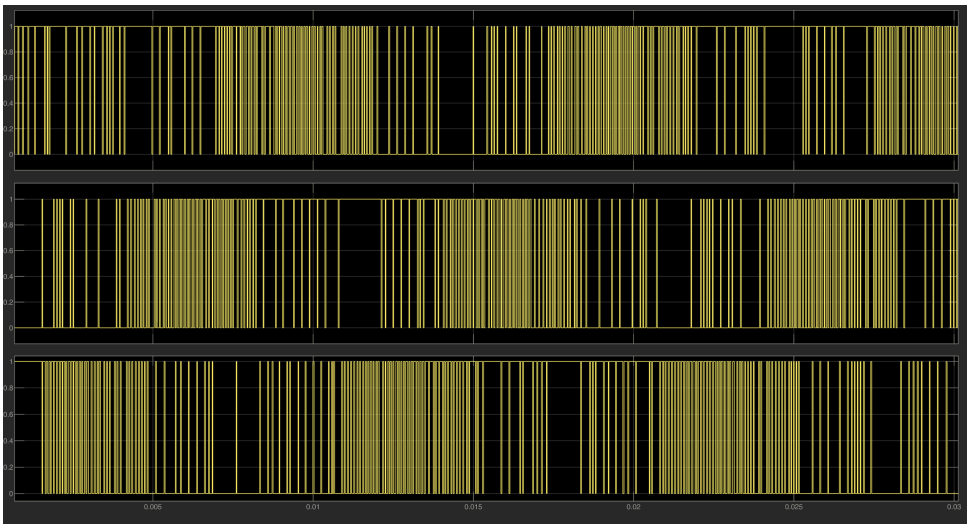
    % Generate 3-phase reference currents
    Ia = Im * sin(wt - phi);
    Ib = Im * sin(wt - 2*pi/3 - phi);
    Ic = Im * sin(wt + 2*pi/3 - phi);

    Iabc_ref = [Ia; Ib; Ic];
end
```

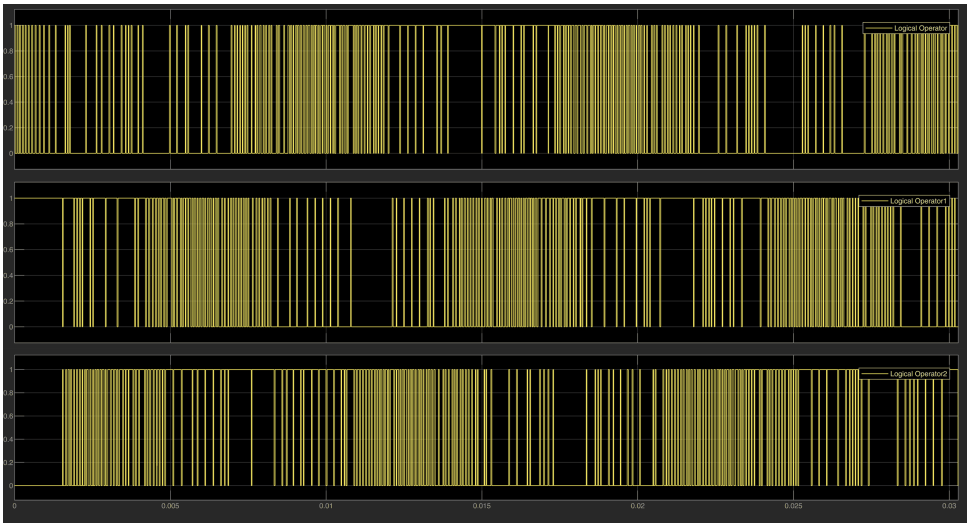
2.2 Hysteresis Current Control and Gate Signals

The generated reference currents are continuously compared with the actual measured grid currents. The resulting error signal, $e(t) = i^*(t) - i(t)$, is fed into Relay blocks with a predefined tolerance band ($\pm\Delta i$).

When the actual current hits the upper limit of the hysteresis band, the upper IGBT of the corresponding leg is turned off, and the lower IGBT is turned on, forcing the current to decrease. Conversely, when it hits the lower limit, the switching state is reversed. This logic generates the switching pulses for the upper IGBTs. Logical NOT operators generate the complementary pulses for the lower IGBTs to avoid short-circuiting the DC link. While this method results in a variable switching frequency, it tightly bounds the current error, yielding excellent tracking capability.



(a) Upper IGBTs Switching Signals



(b) Lower IGBTs Complementary Switching Signals

Figure 2: Complementary PWM Gate Signals generated by the Hysteresis Controller.

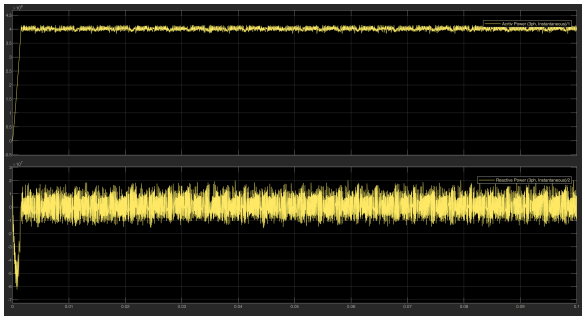
3 Simulation Results and Analysis

To thoroughly evaluate the performance of the implemented VSC and its hysteresis controller, the system was subjected to five distinct operational cases, covering steady-state operations at various power factors and a severe dynamic transient.

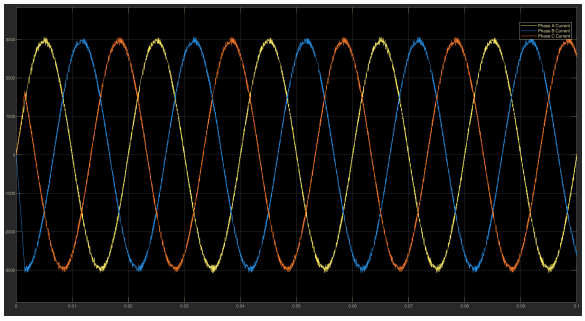
3.1 Case 1: Injection of +405 MW and 0 MVAR

In this baseline scenario, the VSC operates at unity power factor, injecting purely active power into the grid. The reactive power is strictly maintained at zero, which is the ideal condition to minimize transmission line losses and maximize the active power transfer capacity of the grid infrastructure.

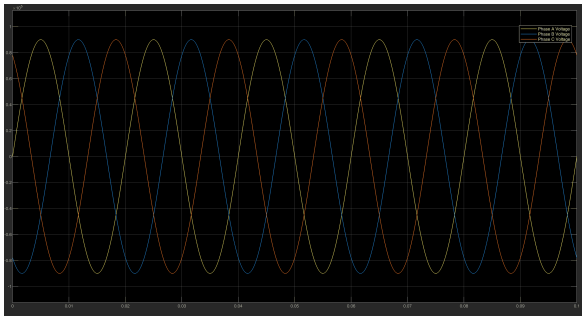
As observed in the waveforms, the grid current is in phase with the grid voltage, verifying the unity power factor operation. Based on the theoretical calculations, the peak phase current is expected to be 3000 A $\left(I_m = \frac{2}{3} \times \frac{405 \text{ MVA}}{90 \text{ kV}}\right)$, which matches the simulation output. The inverter phase voltage exhibits a clean two-level PWM characteristic, confirming proper switching action by the Hysteresis controller as it successfully bounds the actual current around the sinusoidal reference.



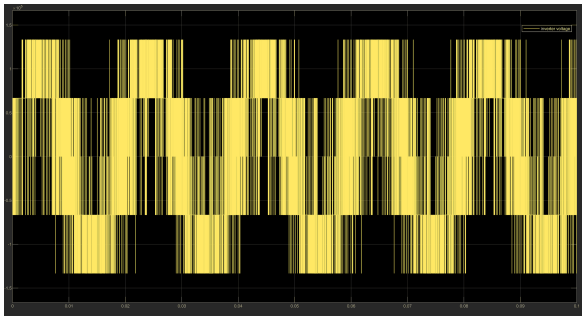
(a) Active and Reactive Power



(b) Grid Currents



(c) Grid Voltages



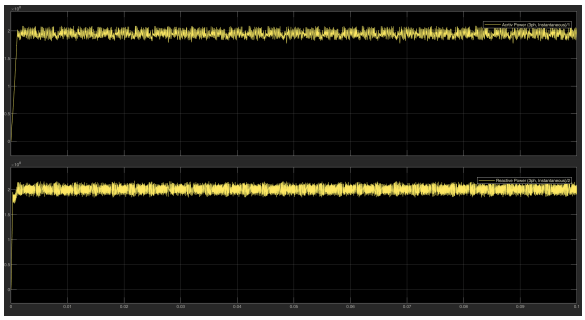
(d) Inverter Phase Voltage

Figure 3: Simulation waveforms for Case 1 (Unity Power Factor).

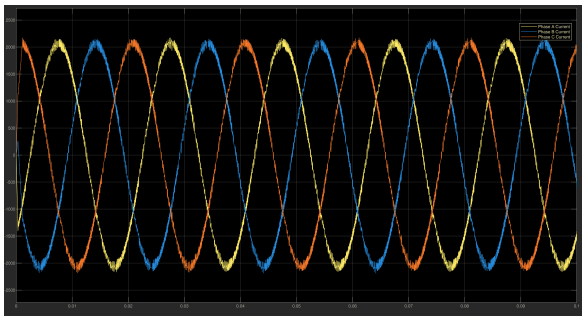
3.2 Case 2: Injection of +200 MW and +200 MVAR

The system is commanded to deliver both active and reactive power symmetrically ($P = 200$ MW, $Q = 200$ MVAR). This case demonstrates the VSC's dual capability to act as an active power source while simultaneously functioning as a reactive power compensator to support local grid voltage profiles.

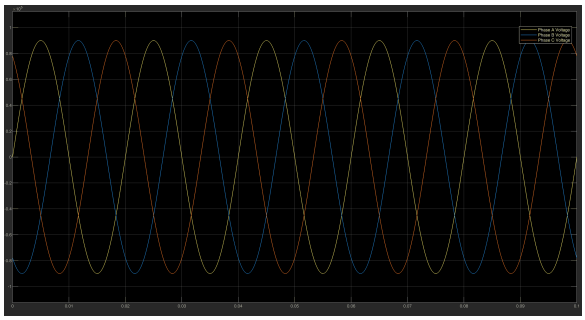
Analytically, this equal distribution of real and reactive power results in a power factor angle of 45°. This theoretical angle is clearly visible as a precise phase displacement between the grid voltage and the injected current waveforms in the simulation. The apparent power required from the converter is $S = \sqrt{200^2 + 200^2} \approx 282.8$ MVA. This dictates a theoretical peak current of approximately 2095 A, which the hysteresis controller tracks accurately without exceeding the designated switching band.



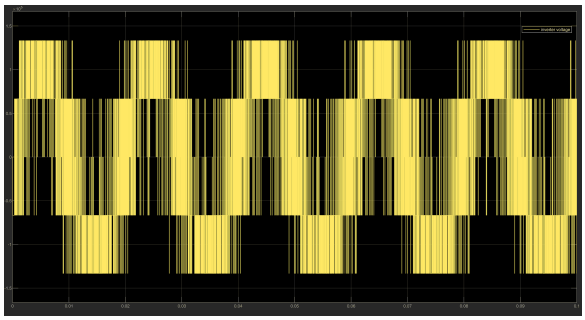
(a) Active and Reactive Power



(b) Grid Currents



(c) Grid Voltages



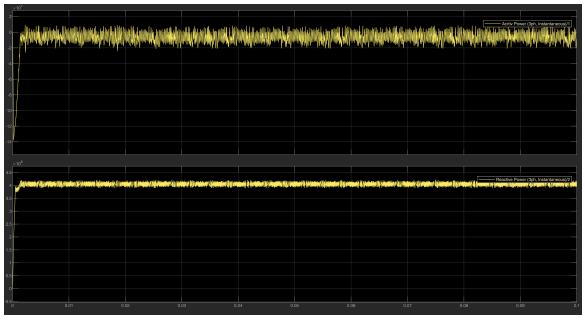
(d) Inverter Phase Voltage

Figure 4: Simulation waveforms for Case 2 ($P = Q$ injection).

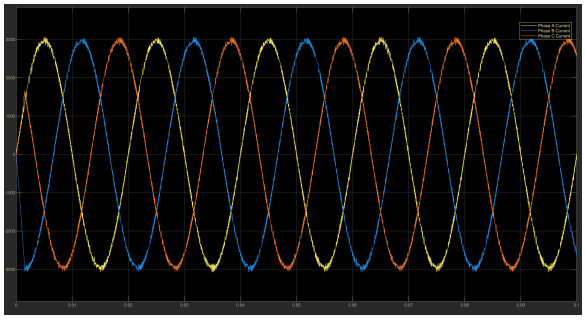
3.3 Case 3: Operation as STATCOM (0 MW, +405 MVAR)

In this case, the active power reference is intentionally set to zero ($P = 0$), forcing the VSC to operate purely as a Static Synchronous Compensator (STATCOM). The converter injects maximum reactive power (+405 MVAR) into the AC network. This operation mode is highly beneficial in modern power systems for providing dynamic voltage regulation and compensating heavily loaded inductive transmission lines.

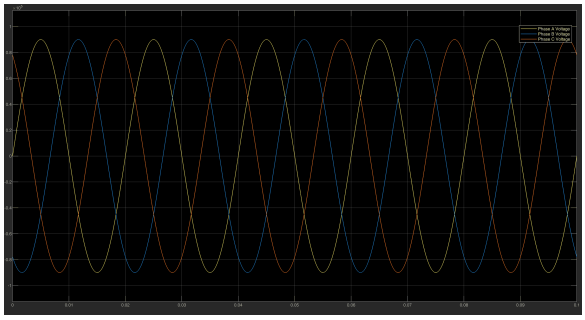
The absence of active power transfer is physically evident in the waveforms, as the grid currents are shifted exactly 90° (leading/lagging) relative to the phase voltages. During this mode, the DC link capacitor theoretically exchanges no net active power with the AC grid (ignoring internal switching losses and filter resistance). The controller successfully enforces the zero active power condition while strictly maintaining the 405 MVAR reactive power injection.



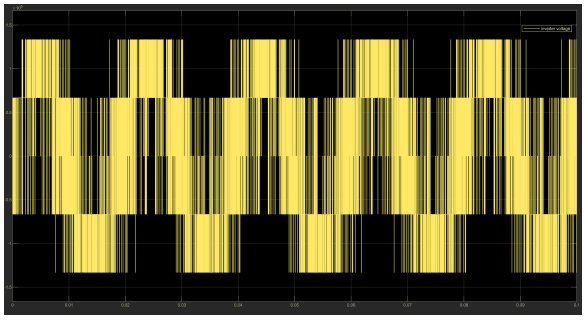
(a) Active and Reactive Power



(b) Grid Currents



(c) Grid Voltages



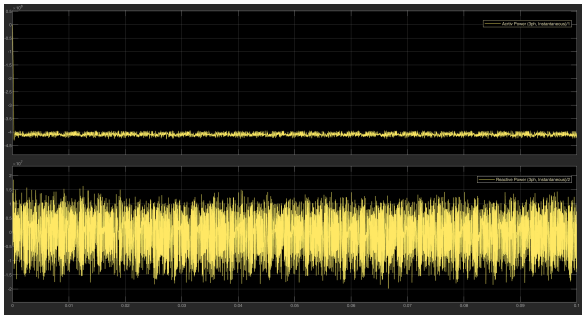
(d) Inverter Phase Voltage

Figure 5: Simulation waveforms for Case 3 (STATCOM Mode).

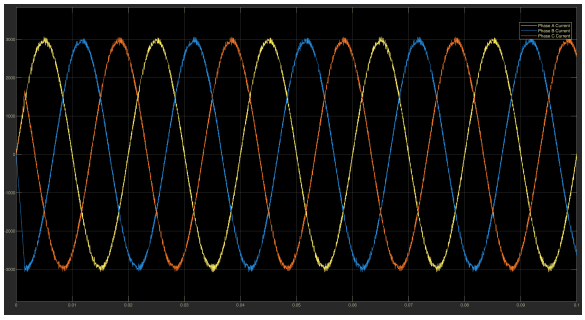
3.4 Case 4: Negative Active Power Flow (-405 MW, 0 MVAR)

This scenario tests the bidirectional power flow capability of the VSC, a crucial feature for multi-terminal HVDC grids and regenerative braking applications. With a negative active power reference ($P = -405$ MW), the converter transitions into rectification mode, drawing active power from the AC grid and delivering it to the DC side.

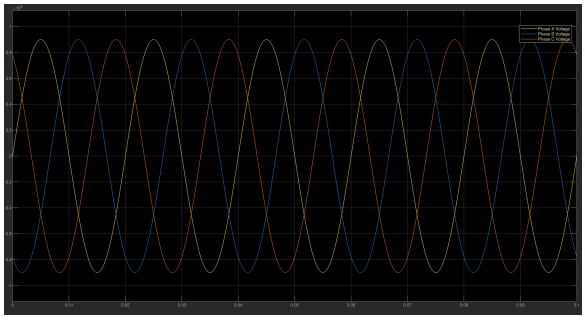
The magnitude of the current remains constant at 3000 A since the absolute apparent power requirement is unchanged from Case 1. However, the waveforms reveal the underlying mechanism of power reversal: a complete 180° phase shift. The grid currents are now operating in anti-phase relative to the grid voltages. This 180° shift is successfully executed by the controller while keeping the reactive power strictly regulated at zero.



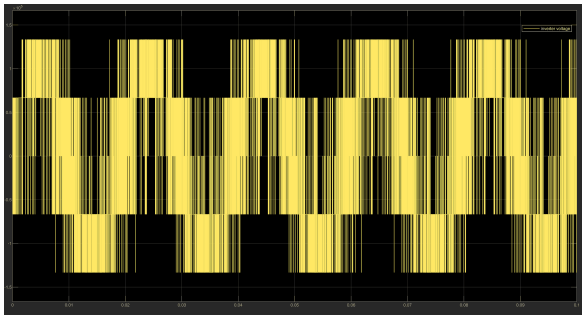
(a) Active and Reactive Power



(b) Grid Currents



(c) Grid Voltages



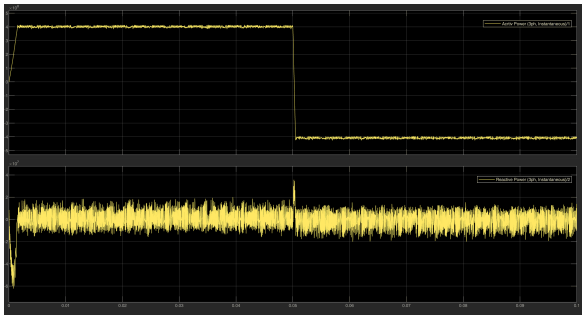
(d) Inverter Phase Voltage

Figure 6: Simulation waveforms for Case 4 (Rectification Mode).

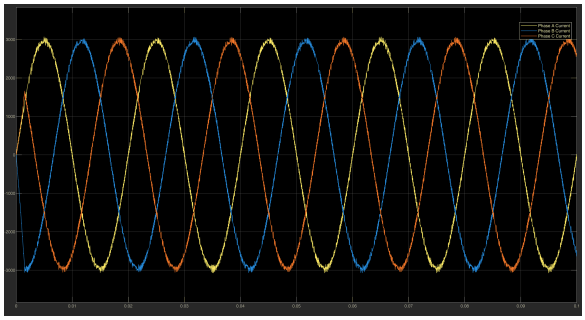
3.5 Case 5: Dynamic Step Change (+405 MW to -405 MW)

The final case investigates the controller's dynamic transient response and stability limits. A severe step change is applied to the active power reference at $t = 0.05$ seconds, commanding an instantaneous transition from full inverter mode (+405 MW) to full rectifier mode (−405 MW).

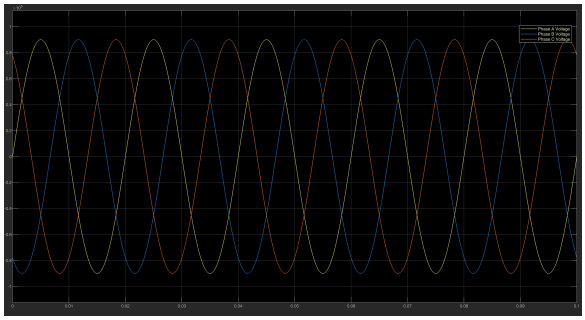
The hysteresis controller tracks the reversal quickly: the active power flow reverses within a fraction of an AC cycle. This is visually corroborated by the grid current waveform, which abruptly shifts its phase by 180° at $t = 0.05$ s. Furthermore, this severe active power transient occurs without inducing significant overshoots or prolonged oscillatory transients in the reactive power profile, which is consistent with decoupled regulation of active and reactive power inherent to this VSC control scheme.



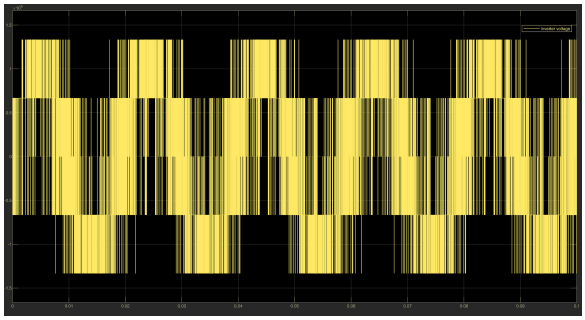
(a) Active and Reactive Power (Dynamic Response)



(b) Current Reversal at $t = 0.05$ s



(c) Grid Voltages



(d) Inverter Phase Voltage

Figure 7: Simulation waveforms for Case 5 (Severe Step Change).

4 Conclusion

The comprehensive simulation results confirm the effectiveness and accuracy of the Two-Level VSC utilizing Hysteresis Current Control. The mathematical generation of reference currents gave accurate current references. The system successfully tracked all steady-state active and reactive power references independently (Cases 1-4) while maintaining the current tightly within the specified hysteresis band. Furthermore, the controller completed the power reversal within a fraction of an AC cycle (Case 5), which supports its use as a basis for VSC-HVDC studies.